MVP Simulator: MDP Formulation, Contextual Bandits, and Benchmark Results

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1 MDP Formulation

The MVP simulator implements a finite-horizon MDP $(\mathcal{S}, \mathcal{A}, P, r, \gamma, T)$:

1.1 State Space

$$\mathcal{S} = \{\texttt{sedentary},\ \texttt{active}\}$$

Two activity states with no additional time-of-day dimension. Time-dependent reward scaling is handled via `reward_multiplier_by_step` (see §1.4 below).

1.2 Action Space

$$\mathcal{A} = \{\texttt{nudge},\ \texttt{idle}\}$$

A binary intervention: send a motivational message (`nudge`) or do not (`idle`).

1.3 Transition Function

$P(s' \mid s, a)$ is a configurable $2 \times 2 \times 2$ tensor. Rows index current state $s \in \{\text{sedentary}, \text{active}\}$ and columns index next state s' in the same order:

$$P_{\text{nudge}} = \begin{bmatrix} 0.7 & 0.3 \\ 0.5 & 0.5 \end{bmatrix} \qquad P_{\text{idle}} = \begin{bmatrix} 0.9 & 0.1 \\ 0.4 & 0.6 \end{bmatrix}$$

Sending a reminder increases $P(\text{active} \mid \text{sedentary})$ from 0.1 to 0.3 but decreases $P(\text{active} \mid \text{active})$ from 0.6 to 0.5, consistent with Reactance theory (reminders can backfire for already-active users).

Optional `reward_multiplier_by_step`: a per-step scaling factor that can zero out rewards at certain daily steps (soft mask). When absent, all multipliers default to 1.0. The per-step reward is computed as base reward $\times$ multiplier$[t]$ where t is the step-of-day index.

1.4 Reward

The base reward is a vector indexed by next state:

$$r = [0.0,\ 1.0]^\top \quad (\text{sedentary},\ \text{active})$$

The per-step reward at time t scales by the multiplier:

$$r(s, a, t) = m[t \bmod S] \cdot r(s')$$

where $S =$ `steps_per_day`, $m \in \mathbb{R}^S$ is `reward_multiplier_by_step` (default $m_i = 1.0$), and s' is the next state after taking action a in state s. Currently $m_i = 1.0$ for all i, so $r(s, a, t) = r(s')$.

Single-timescale placeholder. Multi-timescale reward (immediate + delayed body measure) is deferred to Phase 2.

1.5 Episode Structure

$T =$ `episode_days` $\times$ `steps_per_day` timesteps. Default: $T = 90$ days $\times$ 5 steps/day $= 450$.

2 Baseline Agents

All agents in this section are state-blind bandits: they do not condition on the current activity state. From the transition matrix, $E[r \mid \text{nudge}] \approx 0.375$, $E[r \mid \text{idle}] = 0.20$, and a state-aware policy that nudges only when sedentary would achieve ≈ 0.429. This gap motivates the contextual agents in §4.

2.1 Thompson Sampling

Beta-Bernoulli TS with prior $\text{Beta}(1, 1)$ per action. Posterior update:

$$\alpha_a \leftarrow \alpha_a + \mathbb{1}[s'.\texttt{activity} = \texttt{active}], \quad \beta_a \leftarrow \beta_a + \mathbb{1}[s'.\texttt{activity} = \texttt{sedentary}]$$

Action selected by sampling $\theta_a \sim \text{Beta}(\alpha_a, \beta_a)$ and choosing $a^* = \arg\max_a \theta_a$.

2.2 Epsilon-Greedy

$\epsilon = 0.1$. With probability ϵ, select uniformly at random; otherwise select $\arg\max_a Q(a)$. Q-values updated via incremental average:

$$Q(a) \leftarrow Q(a) + \frac{r_t - Q(a)}{n_a}$$

2.3 UCB1

Upper Confidence Bound with exploration parameter $c = 2.0$. Selects:

$$a^* = \arg\max_a \left[Q(a) + c\sqrt{\frac{\ln N}{N_a}} \right]$$

where N is total steps and N_a is the number of times action a was chosen. The confidence bonus shrinks as N_a grows, naturally shifting from exploration to exploitation. Each action is pulled once before UCB is applied.

2.4 Random Baseline

Uniform random action selection. No learning. Serves as lower bound. The mixture chain $P_{\text{random}} = \frac{1}{2}(P_{\text{nudge}} + P_{\text{idle}})$ gives $E[r] \approx 0.308$ per step, matching the empirical result within one standard deviation.

3 Initial Results

All agents run over 50 random seeds, 450 timesteps each. Results are reported as mean ± standard deviation across seeds.

3.1 Standard Config (`configs/mvp.yaml`)

Four baseline agents (TS, EG, UCB, Random). Agent definitions and hyperparameters live in `docs/mvp/configs/mvp.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	159.6 ± 12.7	0.355	0.373
Epsilon-Greedy ($\epsilon = 0.1$)	158.1 ± 17.8	0.351	0.370
UCB ($c = 2.0$)	148.4 ± 12.6	0.330	0.354
Random	137.3 ± 12.3	0.305	0.319

Table 1: Results on `configs/mvp.yaml` (50 seeds, 450 timesteps). TS and ϵ-greedy perform similarly; TS has lower variance. UCB ($c = 2.0$) underperforms relative to the tuned value ($c = 0.5$, see §6).

3.2 Masked Config (`docs/mvp/configs/mvp_masked.yaml`)

`reward_multiplier_by_step: [1, 1, 1, 1, 0]` – step 4 (end of day) yields zero reward regardless of outcome. Results are 10 seeds.

Agent	Total Reward	Mean Reward/Step
Thompson Sampling ($\alpha_0 = \beta_0 = 1$)	126.6 ± 6.8	0.281 ± 0.015
Epsilon-Greedy ($\epsilon = 0.1$)	127.3 ± 10.2	0.283 ± 0.023
UCB ($c = 2.0$)	116.9 ± 8.6	0.260 ± 0.019
Random	107.8 ± 6.7	0.240 ± 0.015

Table 2: Results on `docs/mvp/configs/mvp_masked.yaml` (10 seeds, 450 timesteps). Reward drops ≈ 20% across all agents, matching the 1/5 masked fraction. Relative ordering is preserved.

4 Contextual Bandits

Context $c_t = s_t$.`activity` $\in$ {`sedentary`, `active`}. Each algorithm maintains separate parameters per (c, a) pair:

- **Thompson Sampling:** $(\alpha_{c,a}, \beta_{c,a})$ posteriors. Sample $\theta_{c_t,a} \sim \text{Beta}(\alpha_{c_t,a}, \beta_{c_t,a})$, pick $a^* = \arg\max_a \theta_{c_t,a}$.
- **Epsilon-Greedy:** $Q_{c,a}$ values. With probability ϵ explore, otherwise $a^* = \arg\max_a Q_{c_t,a}$.
- **UCB:** $a^* = \arg\max_a \left[Q_{c_t,a} + c\sqrt{\ln(N_{c_t})/n_{c_t,a}}\right]$ where $N_{c_t} = \sum_a n_{c_t,a}$.
- **Decaying Epsilon-Greedy:** Linear decay: $\epsilon_t = \max(\epsilon_{\min},\ \epsilon_{\text{start}}(1 - t/T_{\text{decay}}))$.

4.1 Optimal Policy

Nudge when sedentary ($0.3 > 0.1$), idle when active ($0.6 > 0.5$). Contextual bound: $E[r] = 3/7 \approx 0.429$ per step, 192.9 total (14.3% above non-contextual optimal).

5 Hyperparameter Optimisation

Grid search over key parameters (50 seeds, 450 timesteps, both standard and contextual variants):

$$\begin{aligned} \text{EG:} \quad & \epsilon \in [0.01, 0.50],\ \Delta 0.05 \\ \text{D-EG:} \quad & \epsilon_{\text{start}} \in [0.1, 1.0],\ \Delta 0.1; \quad T_{\text{decay}} \in [25, 700],\ \Delta 25 \\ \text{UCB:} \quad & c \in [0.1, 10.0],\ \Delta 0.5 \end{aligned}$$

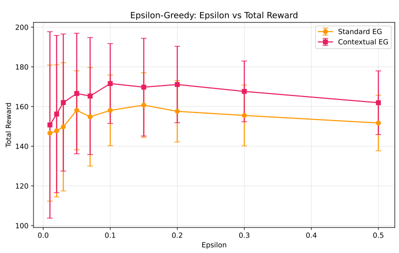

Figure 1: *
EG: ϵ vs reward

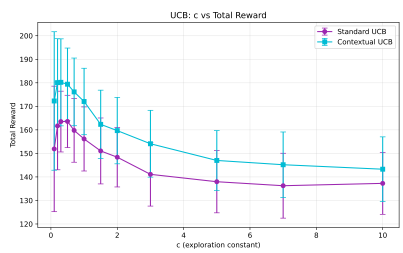

Figure 2: *
UCB: c vs reward

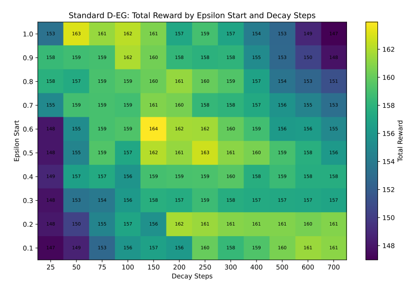

Figure 3: *
D-EG: heatmap

Figure 4: Hyperparameter sensitivity (50 seeds, ± 1 SD). EG optimum $\epsilon = 0.15$, UCB optimum $c = 0.5$ (both interior). D-EG total reward plateaus above $\epsilon_{\text{start}} = 0.2$; $\epsilon_{\text{start}} = 1.0$ is the theoretical maximum but $\epsilon_{\text{start}} = 0.2$ is chosen in benchmarks for lower variance.

$\epsilon = 0.05$ vs $\epsilon = 0.10$: identical means (158.1), but $\epsilon = 0.10$ has 11% lower variance.

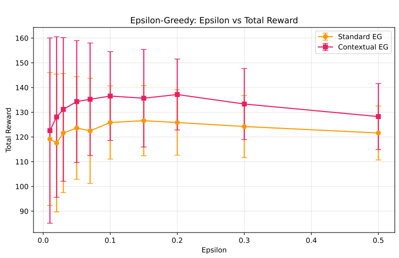

Figure 5: *
EG (masked): ϵ vs reward

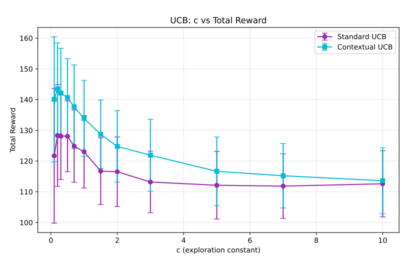

Figure 6: *
UCB (masked): c vs reward

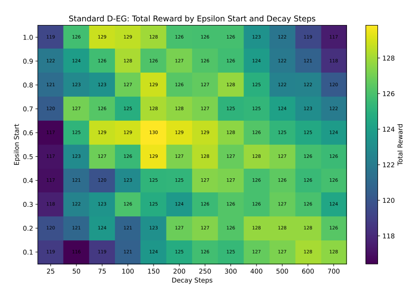

Figure 7: *
D-EG (masked): heatmap

Figure 8: Hyperparameter sensitivity under masked reward (50 seeds, ± 1 SD). Optima are unchanged from the non-masked case; the 20% reward floor shift is uniform across all hyperparameter settings.

6 Benchmark Results

50 seeds, 450 timesteps. Optimised hyperparameters: EG $\epsilon = 0.05$, D-EG $\epsilon_{\text{start}} = 0.2$, $T_{\text{decay}} = 200$, $\epsilon_{\text{min}} = 0.01$, UCB $c = 0.5$. All agent definitions live in `docs/mvp/configs/mvp_extensions.yaml`.

Agent	Total Reward	Per Step	Last 50 Steps	vs Contextual Opt
Contextual UCB	179.4 ± 15.4	0.399	0.430	−7.0%
Contextual TS	178.7 ± 15.8	0.397	0.429	−7.4%
Contextual EG	166.6 ± 30.4	0.370	0.405	−13.6%
Contextual D-EG	167.0 ± 35.2	0.371	0.396	−13.4%
Standard UCB	163.6 ± 11.1	0.364	0.382	−15.2%
Standard D-EG	161.7 ± 18.4	0.359	0.375	−16.2%
Standard TS	159.6 ± 12.7	0.355	0.373	−17.2%
Standard EG	158.1 ± 19.9	0.351	0.375	−18.0%
Random	137.3 ± 12.3	0.305	0.319	−28.9%
Contextual optimal	192.9	0.429	0.429	—
Non-contextual optimal	168.8	0.375	0.375	—

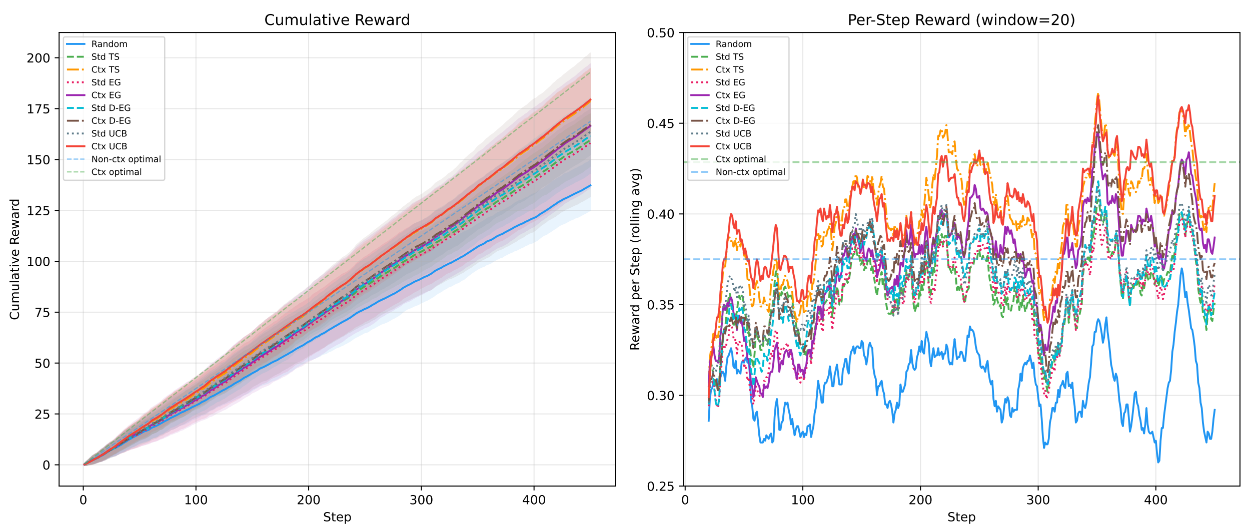

Figure 9: Cumulative reward and per-step reward (50 seeds, ± 1 SD). Contextual agents converge toward $3/7 \approx 0.429$; standard agents plateau lower.

Contextual TS and UCB reach the per-context optimal (0.429) by the last 50 steps. Standard UCB benefits most from hyperparameter search (from −23.1% with $c = 2.0$ to −15.2% with $c = 0.5$). D-EG front-loads exploration, achieving 161.7 vs EG's 158.1 in the standard setting; the contextual advantage disappears as per-context Q-values already provide sufficient exploration. Random (0.305) anchors the lower bound, 28.9% below the contextual optimum.

7 Masked Reward Benchmark

The same 9-agent benchmark on `docs/mvp/configs/mvp_masked.yaml` where step 4 (end of day) is zero-reward. Results are 50 seeds.

Agent	Total Reward	Per Step	Last 50 Steps	vs Contextual Opt
Contextual TS	141.9 ± 13.0	0.315	0.336	0.0%
Contextual UCB	140.7 ± 12.7	0.313	0.334	0.0%
Contextual D-EG	135.7 ± 26.8	0.302	0.323	−12.1%
Contextual EG	134.3 ± 24.7	0.298	0.321	−13.0%
Standard UCB	128.1 ± 11.5	0.285	0.300	−17.0%
Standard D-EG	126.8 ± 19.4	0.282	0.293	−17.8%
Standard TS	125.8 ± 10.9	0.280	0.289	−18.5%
Standard EG	123.6 ± 20.8	0.275	0.292	−19.9%
Random	110.0 ± 10.2	0.244	0.250	−28.7%
Contextual optimal	154.3	0.343	0.343	—
Non-contextual optimal	135.0	0.300	0.300	—

Table 3: Results on `docs/mvp/configs/mvp_masked.yaml` (50 seeds, 450 timesteps, step 4 masked). Relative ordering matches the non-masked benchmark; contextual TS and UCB are tied at the top.

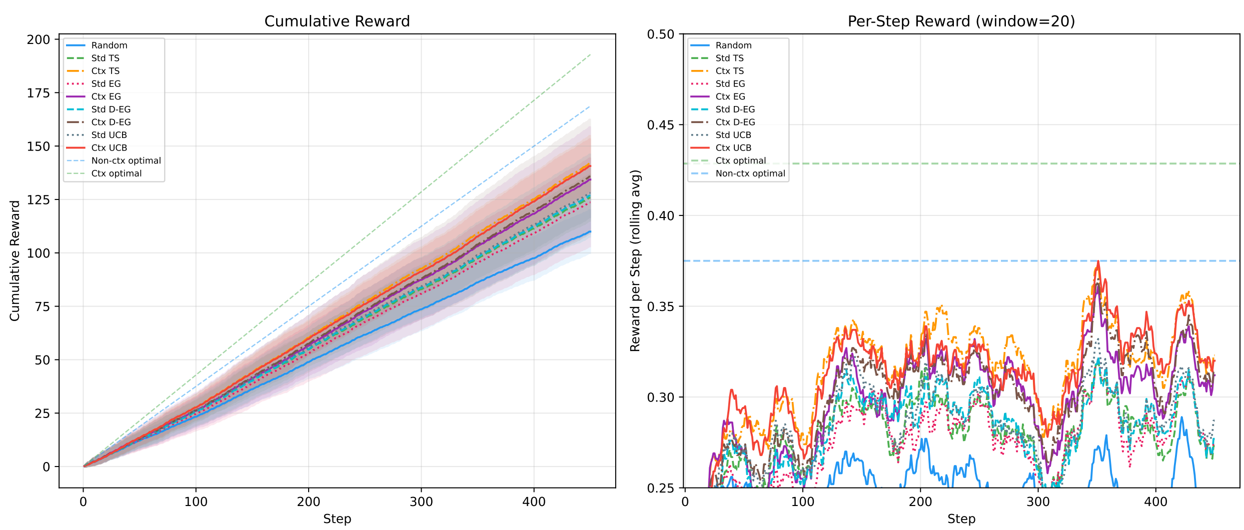

Figure 10: Masked cumulative reward and per-step reward (50 seeds, ±1 SD). Reward floor drops by 20% vs non-masked, but relative ordering is preserved.

8 Known Limitations

- Agents are contextual bandits, not state-aware RL. Full state-aware RL (with state history and temporal-difference learning) is Phase 2.
- Single-timescale reward only. Multi-timescale (immediate + 3-week delayed) is Phase 2.

- No user response model. Burden accumulation and 4 archetypes are Phase 2.
- Transition matrix is fixed and known. Learned transitions from real data are Phase 2.
- Transition probabilities are placeholder values informed by Reactance theory. Calibration against real data is Phase 2.